

Title: Challenge 1 - A recommender system for online shops

Description:

You are a data scientist working for a company. The current task is to build a recommender system for an online marketplace. The recommender system is basically a python code that suggests a list of five top books on “machine learning”. The criteria used by this recommender system includes year of publication (the more recent, the better), number of editions (the higher the better), and useful for “python”.

In reality, you, as a data scientist, should opt for retrieving real data from online platforms/shops. Since this is the first challenge, it is decided to start from a simple task. Thus, a real dataset from <https://openlibrary.org> was obtained and passed to you as the main dataset required in this challenge for developing the recommender system.

The dataset is in CSV format. It includes a list of 500 books. Since the data is real, it may include books not correctly related to the machine learning topic even though the keyword “machine learning” was used for collecting this dataset. In addition, the dataset may have missing information, duplicated entries, book names in other languages than English that should be removed. Thus, before developing the recommender system, it is advised to clean the data.

The important factors for recommending a book is to emphasize on when the book has been published. As the domain of machine learning is evolving fast, therefore the books published in recent years can be assigned higher scores. The second important factor is the number of editions, which can be a good indicator of usefulness. Furthermore, the user would like to read about the machine learning techniques and employ such models in “python”. Therefore, including “python” in the book “title” or “subjects” or “description” can be a better option. Considering these factors, each student can develop a self-defined but justified ranking system to develop a function for recommending which book to buy. The recommender system should be tested for 3, 5, and 10 top books, while the recommended results should be in line with the requirements.

For improving the recommender system, students may enhance the main dataset by adding further information or books to the dataset. The new information can be gathered from online platforms. This is totally optional but has further points in the evaluation.

The students need to write a short report (maximum two pages) and contain the first 5 top books suggested by the code. Based on the performance of the final recommender function, the report can also describe the function used for scoring books, how the code works and clean the original dataset, and provide a short discussion of limitations and future improvements of their developed python codes.